

Wire Specifications:
 16AWG 1.36 mm² Spec44
 22AWG 0.32mm² Spec44

LV Battery

- LV-BMS SSR High-Side Switch
- 30A Fuse
- LV_BATT_ON
- LV Battery 12V

Emergency Buttons

- R ROLL HOOP ES-2012-T112 10.1A
- L ROLL HOOP ES-2012-T112 10.1A
- BOTS RS PRO 448-0747 5A
- COCKPIT ES-2021-T111 10.1A

BSPD

- BSPD_RLY G5Q-1A-EU DC12
- BSPD_ERR A1
- 16.7mA
- Error resets on manual LVMS power cycle or after >10s automatically if opening condition no longer present

INERTIA SW

- Sensata Resettable Crash Sensor
- 10A

Accumulator

- TRACTIVE RELAY CONTROLLER PCB**
 - Controls precharge sequence and closing of AIRs
 - Prevents re-closing of AIRs if SDC closes without power cycle
 - Sends TS READY to and awaits TS ON signal from VCU
 - Powers AIR Coils directly from SDC
- CLOSE -AIR EV200HAANA 156mA
- CLOSE +AIR EV200HAANA 156mA
- CLOSE PRECHARGE DCNLEV50-EN 122mA
- 500A
- CELLS 108s5p
- Precharge Resistor HSA50390RJ
- Not connected
- TS OUT HVSL800082XXXX

Discharge from Inverter side

- SDC
- DISCHARGE RLY Pickering 67-1-C-12/5
- Discharge Resistor HSA252K7J
- TS+
- TS-
- 80mA
- 3A

AIRS:

- 1.7W @ 12V DC hold power
- 156mA coil current
- Total 312mA

Total SDC Consumption = AIR Coil Current + Discharge Coil current = 312mA + 80mA = 392mA

All components in SDC rated to handle current

AMS/IMD Latching PCB

- (PCB integrated with TSAL Logic)
- AMS RLY G5Q-1A-EU DC12
- IMD RLY G5Q-1A-EU DC12
- 16.7mA
- Hardware latching via discrete logic
- Reset button located out of reach of driver
- AMS_FLT
- IMD_FLT

Other Components:

- TSMS Littelfuse 08099080 150A
- HVD INTERLOCK Hirose EM30MSD 1A
- TSAC CONNECTOR INTERLOCK HVSL800082XXXX 17A
- Orion BMS 2 AMS
- Bender IR155-3203 IMD

Vehicle: STAG 12
 Drawn By: Adam Eastman
 Checked By: John McGarvey, Ray Wang
 CAD Part: N/A
SUFST
 Sheet: Shutdown Loop
 File: shutdown_loop.kicad_sch

Title: Shutdown Circuit	
Size: A3	Date: 2026-03-23
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